

STANDARD OPERATING PROCEDURE (SOP)
Implementation of IEC 60870-5-104 Ethernet to OPC UA Conversion
Rockwell Automation OptixEdge 2800E-ANA-S

Document Control

Document Title	IEC 60870-5-104 to OPC UA Conversion SOP
Equipment	Rockwell Automation OptixEdge 2800E- ANA-S
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Prepared By	Engineering Team
Reviewed By	_____
Approved By	_____
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1. Purpose

This SOP describes the complete procedure to implement IEC 60870-5-104 Ethernet to OPC UA protocol conversion using Node-RED and Compact OPC UA Server on Rockwell Automation OptixEdge 2800E-ANA-S.

2. Scope

This procedure covers Portainer IO access, Node-RED deployment, firewall configuration, IEC104 connection, Node-RED flow import, Compact OPC UA Server configuration, certificate generation, namespace setup, multi-tag expansion, testing, troubleshooting, and validation.

3. System Architecture

IEC104 Server: 192.168.2.113:2404

Optix Edge IP: 192.168.2.10

Node-RED: http://192.168.2.10:1880

Portainer IO: https://192.168.2.10:9443

OPC UA Endpoint: opc.tcp://192.168.2.10:4840/UA/NodeRED/Compact

4. Prerequisites

- Rockwell Automation OptixEdge 2800E-ANA-S
- Portainer administrator account
- Node-RED Docker container
- IEC104 Server reachable at 192.168.2.113:2404
- Flow file Edge_IEC104 OPCUA.json
- Network access between IEC Server and Optix Edge

Data Flow:

IEC104 Server → Node-RED IEC104 Master → Tag Mapping → OPC UA Compact Server
→ OPC UA Client/SCADA

5. Safety / Cybersecurity

- Use strong Portainer and Node-RED credentials.
- Allow only required ports (2404, 1880, 4840).
- Avoid exposing OPC UA directly to internet.
- Back up Node-RED flow before modifications.

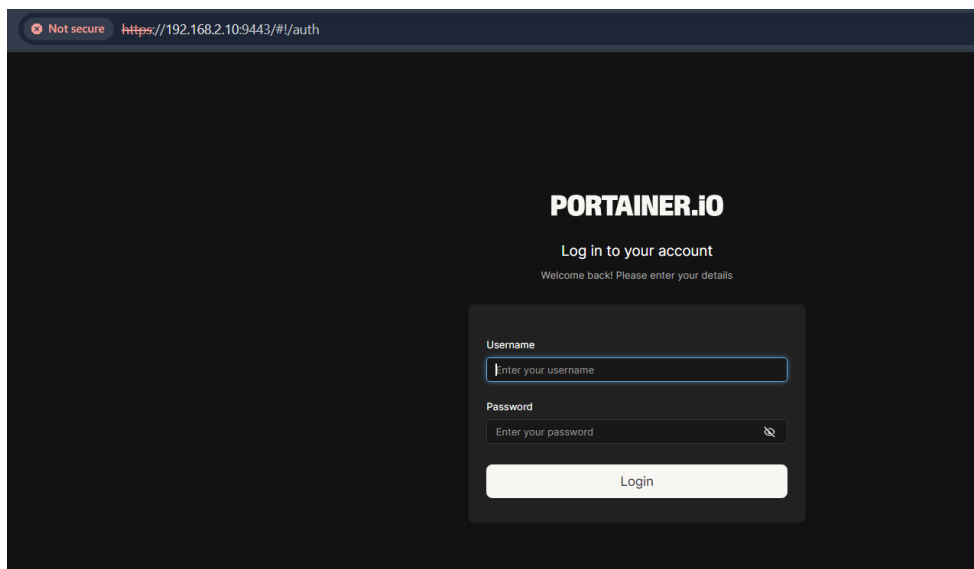
6. Firewall Configuration

Allow IEC104 communication between IEC Server and Optix Edge.

1. Temporarily disable firewall for testing if required.
2. Create inbound allow rule for TCP Port 2404.
3. Create outbound allow rule for TCP Port 2404.
4. Verify communication between 192.168.2.113 and 192.168.2.10.

7. Open Portainer IO

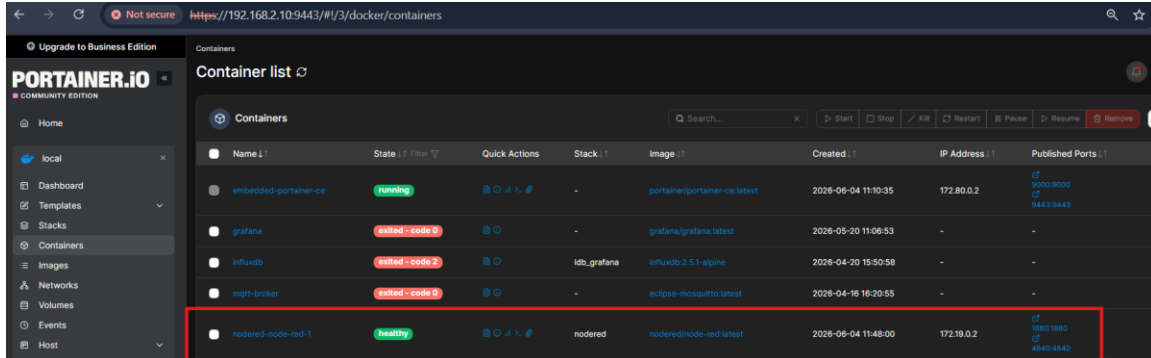
Open browser: <https://192.168.2.10:9443>



Login using administrator username and password.

8. Deploy Node-RED Container

Deploy Node-RED stack in Portainer.



Optional: Stack -WEB Editor to pull the image & container of node red :

version: '3'

services:

nodered:

image: nodered/node-red:latest

restart: unless-stopped

ports:

- "1880:1880"

- "4840:4840"

volumes:

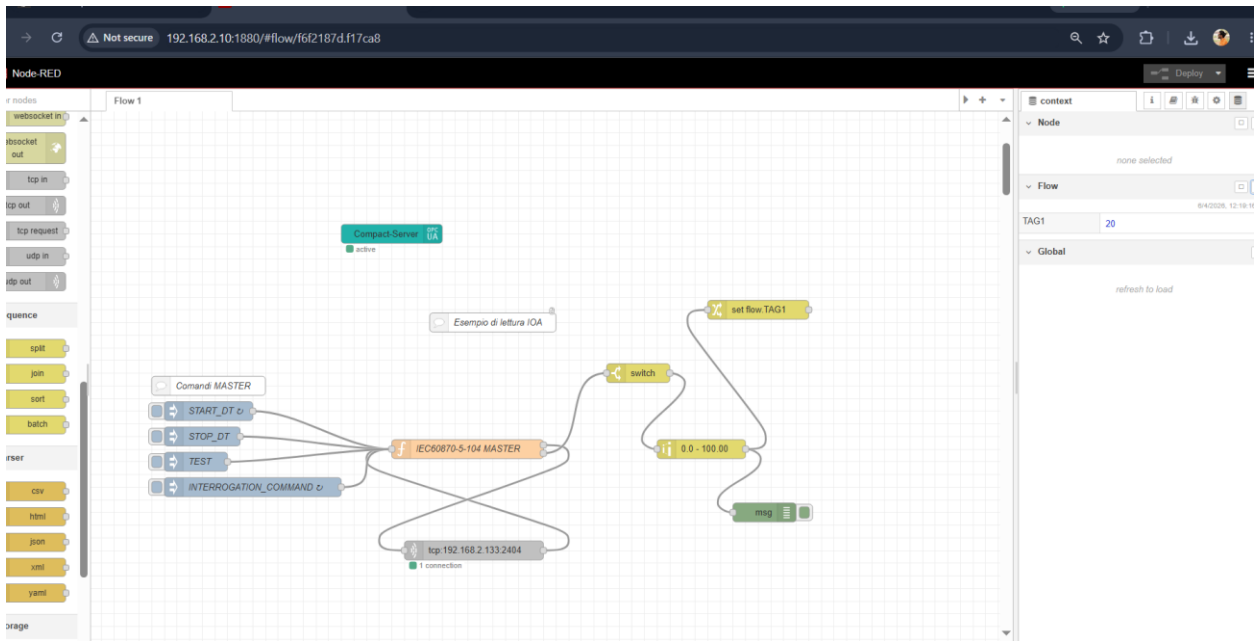
- nodered_data:/data

volumes:

nodered_data:

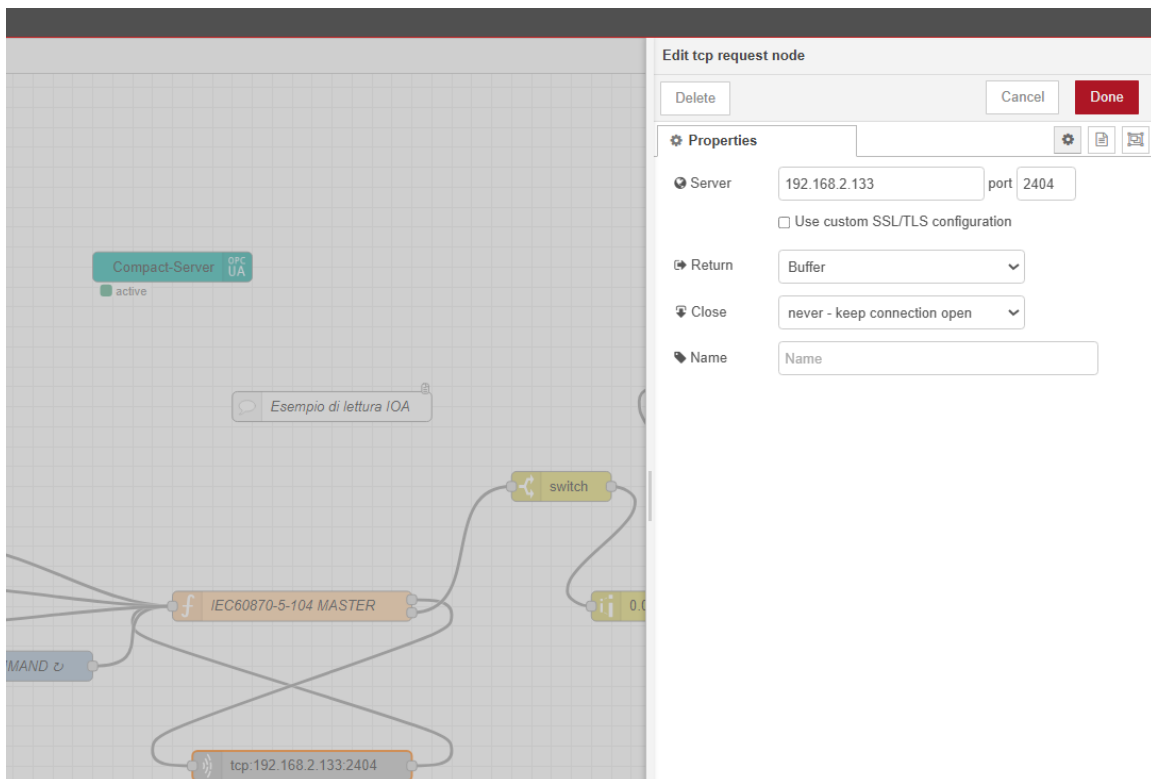
9. Import Node-RED Project

5. Open Node-RED: <http://192.168.2.10:1880>
6. Menu → Import
7. Select Edge_IEC104 OPCUA.json
8. Deploy the flow



10. Modify IEC104 Server IP

Open TCP Request Node and change:

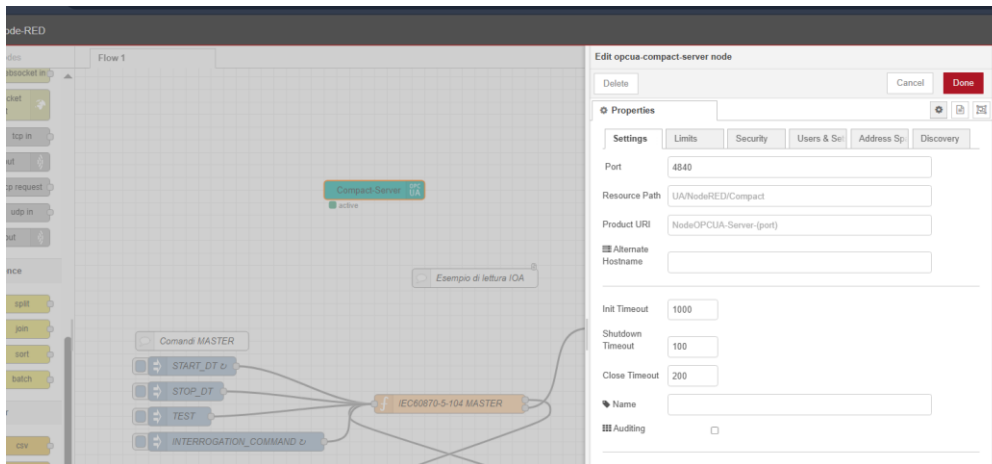


Current: 192.168.2.133:2404

New: IEC Server IP:2404

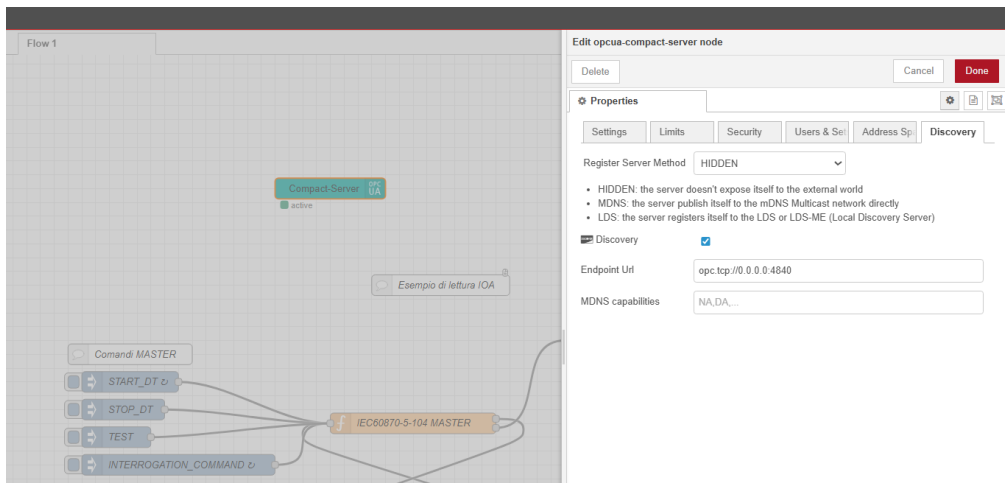
11. Compact OPC UA Server Configuration

Port: 4840



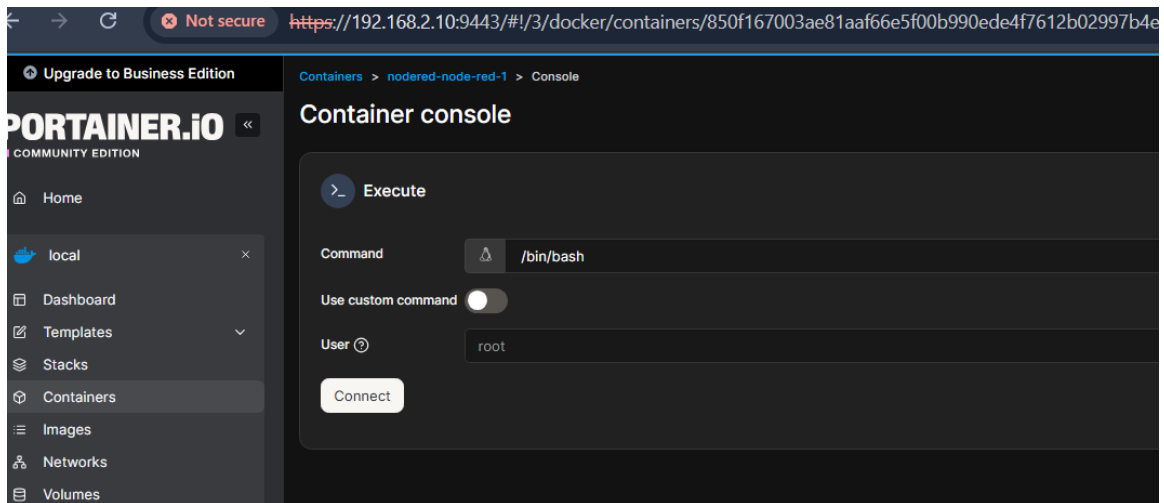
Resource Path: UA/NodeRED/Compact

Endpoint URL: opc.tcp://0.0.0.0:4840



12. Certificate Generation (Run inside Portainer Console) (Important)

Portainer → Containers → nodered-node-red-1 → Console → /bin/sh → Connect



Create folder

```
mkdir -p /data/node_modules/node-red-contrib-opcua-server/certificates/
```

Generate certificate

```
openssl req -x509 -sha256 -nodes -days 365 -newkey rsa:2048 -keyout
/data/node_modules/node-red-contrib-opcua-
server/certificates/server_selfsigned_cert_2048.pem -out /data/node_modules/node-red-
contrib-opcua-server/certificates/server_selfsigned_cert_2048.pem -subj "/CN=NodeRED-
OPCUA-Server"
```

Generate private key

```
openssl req -x509 -sha256 -nodes -days 365 -newkey rsa:2048 -keyout
/data/node_modules/node-red-contrib-opcua-server/certificates/server_key_2048.pem -
out /data/node_modules/node-red-contrib-opcua-
server/certificates/server_selfsigned_cert_2048.pem -subj "/CN=NodeRED-OPCUA-Server"
```

Verify

```
ls -l /data/node_modules/node-red-contrib-opcua-server/certificates/
```

```
.....+..+...+.....+.....+...+....+...+.....+...+...+.....+.
-----
b62ceaaa69ff:~$ ls -l /data/node_modules/node-red-contrib-opcua-server/certificates/
total 16
drwxr-xr-x   7 node-red node-red    4096 Jun  4 05:54 CA
-rw-r--r--   1 node-red node-red     521 Jun  4 05:54 config.js
-rw-----   1 node-red node-red    1704 Jun  4 06:03 server_key_2048.pem
-rw-----   1 node-red node-red    1143 Jun  4 06:03 server_selfsigned_cert_2048.pem
b62ceaaa69ff:~$
```

Expected Verification Output

```
server_key_2048.pem
server_selfsigned_cert_2048.pem
CA/
config.js
```

13. OPC UA Namespace and Address Space

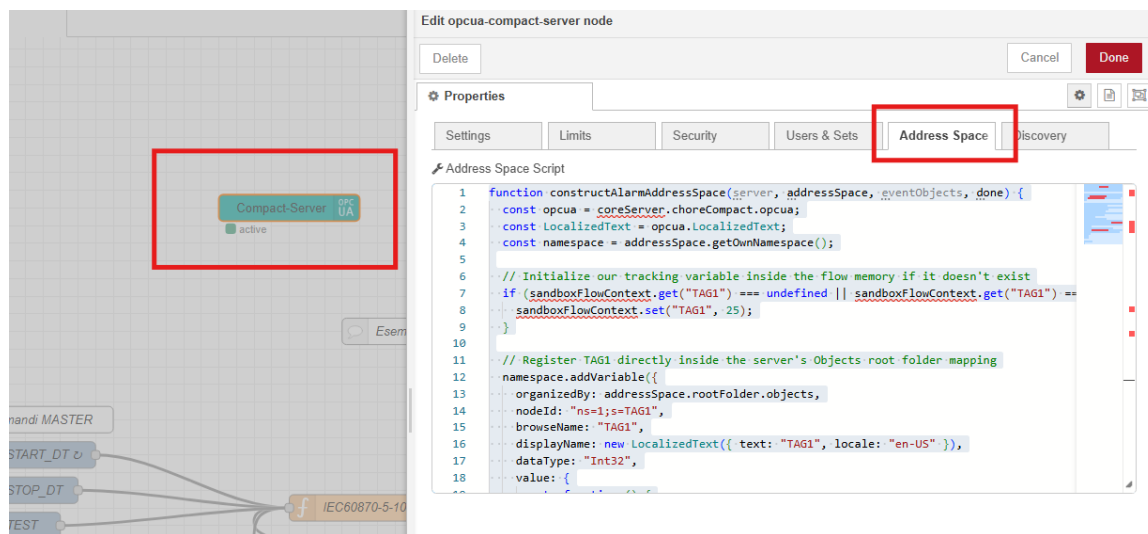
Namespace: ns=1

Node ID: ns=1;s=TAG1

Browse Path: Objects → TAG1

Data Type: Int32

Modify the Script as required:



14. IEC104 MASTER Function Description

START_DT starts data transfer, INTERROGATION_COMMAND requests IEC104 data every 5 seconds, IEC MASTER parses M_ME_TD_1 Type 34 telegrams, extracts IOA, normalizes value and writes to flow.TAG1.

The IEC104 MASTER function performs the following:

Initializes RX/TX sequence counters

Sends START, STOP, TEST, and Interrogation commands

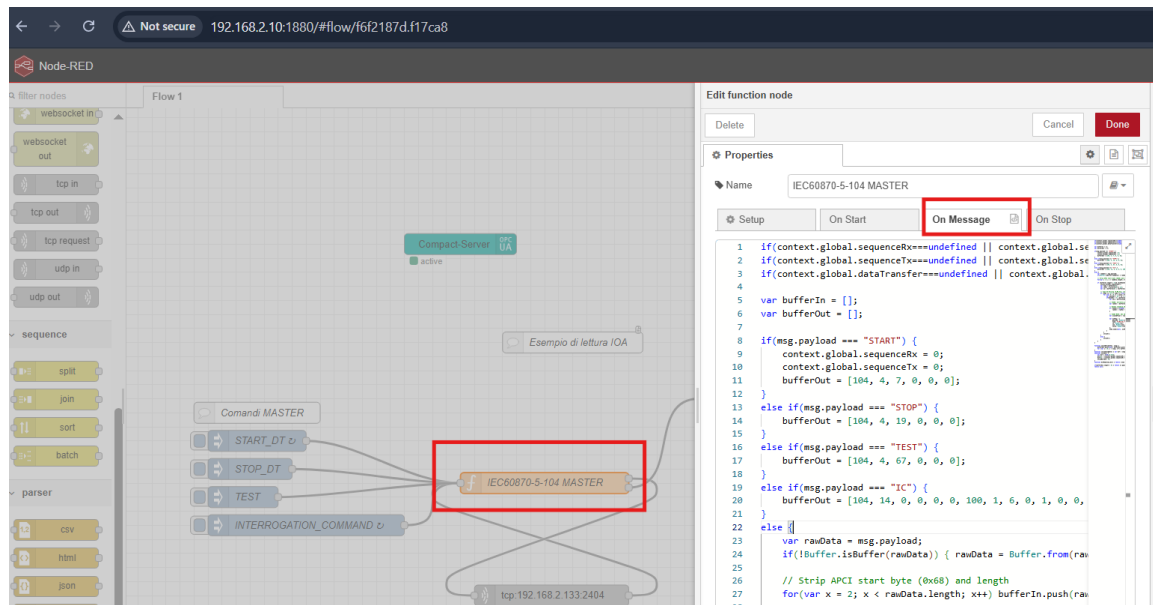
Parses M_ME_TD_1 (Type 34) telegrams

Extracts IOA and normalized analog values

Scales values using IEC104 normalization ($\div 32767$)

Publishes values to OPC UA tags

Add and Modify the On Message in Function node for Multiple Tags to be read by IEC 104 Server.



15. Multi Tag Example (TAG2...TAGN)

Example IOA mapping:

IOA	Tag	NodeID
102	TAG1	ns=1;s=TAG1
103	TAG2	ns=1;s=TAG2
104	TAG3	ns=1;s=TAG3

For additional tags, duplicate namespace.addVariable() and create TAG2, TAG3, ... TAGN.

Modify switch node outputs to map IOA → TAG.

16. Validation & Testing Checklist

- | | | |
|--------------------|------------------------|--------------------------------|
| • Validation Item | • Expected Result | • Status |
| • Portainer Access | • Port 9443 accessible | • ☐ yes |
| • Node-RED Access | • Port 1880 accessible | • ☐ yes |

- IEC104 Connection
- TCP/2404 connected
- ☐ yes
- Compact OPC UA
- Port 4840 active
- ☐ yes
- TAG1 Visibility
- Tag visible in OPC UA browser
- ☐ yes
- Live Data Update
- Tag updates dynamically
- ☐ yes

The screenshot displays the Unified Automation UaExpert interface. The left pane shows the Project tree with 'Servers' and 'Data Access View' expanded. The 'Data Access View' pane shows a table with the following data:

Node ID	Display Name	Value	DataType
NS1(String)TAG1	TAG1	40	Int32

The right pane shows the Simulator window for 'IEC104_SERVER_1'. It contains a table with the following data:

S.No	Common Address	Event Report Type ID	IOA	Value	Quality Bits	Time Stamp
1	1	M_ME_TD_1	100	1.00000	SB BL OV	13:13:28:356 - 04
2	1	M_ME_TD_1	101	0	GD	13:13:28:368 - 04
3	1	M_ME_TD_1	102	0.40000	GD	13:13:28:380 - 04
4	1	M_ME_TD_1	103	0	GD	13:13:28:392 - 04
5	1	M_ME_TD_1	104	0	GD	13:13:28:404 - 04
6	1	M_ME_TD_1	105	0	GD	13:13:28:416 - 04
7	1	M_ME_TD_1	106	0	GD	13:13:28:428 - 04
8	1	M_ME_TD_1	107	0	GD	13:13:28:440 - 04
9	1	M_ME_TD_1	108	0	GD	13:13:28:452 - 04
10	1	M_ME_TD_1	109	0	GD	13:13:28:464 - 04
11	1	C_SE_TA_1	1000	0	GD	13:13:28:476 - 04
12	1	C_SE_TA_1	1001	0	GD	13:13:28:488 - 04
13	1	C_SE_TA_1	1002	0	GD	13:13:28:500 - 04
14	1	C_SE_TA_1	1003	0	GD	13:13:28:512 - 04
15	1	C_SE_TA_1	1004	0	GD	13:13:28:524 - 04
16	1	C_SE_TA_1	1005	0	GD	13:13:28:536 - 04
17	1	C_SE_TA_1	1006	0	GD	13:13:28:548 - 04
18	1	C_SE_TA_1	1007	0	GD	13:13:28:560 - 04
19	1	C_SE_TA_1	1008	0	GD	13:13:28:572 - 04
20	1	C_SE_TA_1	1009	0	GD	13:13:28:584 - 04

The bottom pane shows the Log window with the following entries:

Timestamp	Source	Server	Message
6/4/2026 11:48:...	DA Plugin	@node-red	CreateMonitoredItems succeeded [ret = Good]
6/4/2026 11:48:...	DA Plugin	@node-red	Item [NS1(String)TAG1] succeeded: RevisedSampl...

17. Troubleshooting

- **Cannot connect IEC104:** Check firewall and TCP 2404
- **Compact server inactive:** Verify certificate files exist
- **No OPC UA connection:** Check port 4840 mapping
- **Wrong data:** Verify IOA mapping and scaling

18. Revision History

- | | | | |
|------------|---------------|---------------|---------------|
| • Revision | • Date | • Description | • Author |
| • 1.0 | • 04-Jun-2026 | • SOP | • Engineering |